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Systems-level modeling of cell migration using spatially-resolved single-cell data

Mahdi Pursalim · Kaveh Kavousi · Parisa Shooshtari

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Abstract

Background The ability to capture cellular dynamics at high spatial and molecular resolution is crucial for understanding complex biological processes, including development, tissue remodeling, and disease progression. Single-cell RNA sequencing (scRNA-seq) provides detailed transcriptional profiles of individual cells, while spatial transcriptomics (ST) preserves their spatial organization within tissues. However, integrating these complementary modalities across multiple time points to reconstruct temporal changes in spatial cell-type organization and cell-state composition remains a major challenge. Here, we present a computational framework that integrates time-series scRNA-seq and spatial transcriptomics data to reconstruct spatiotemporal cell-type redistribution and inferred cell-state dynamics.

Results The framework combines reference-based spatial deconvolution with single-cell spatial mapping to generate temporally resolved representations of cellular organization. Temporal changes in spatial cell-type distributions are quantified using complementary abundance-weighted spatial descriptors, whereas inferred cell-state dynamics are characterized through changes in dominant cell-type assignments across consecutive time points. We further evaluated the spatial abundance component of the framework using simulation-based quantitative benchmarking against alternative spatial deconvolution and mapping approaches, using known simulated cell-type compositions as ground truth. We demonstrate the utility and generalizability of the framework using two biologically distinct systems: human glioblastoma organoids (GBOs) and embryonic chicken heart development. Across both datasets, the framework captures temporal patterns of spatial cell-type organization and inferred cell-state composition that are not accessible through static analyses alone.

Conclusions This work provides a broadly applicable computational framework for investigating temporal tissue organization through the integration of single-cell and spatial transcriptomic data, thereby enabling systematic characterization of dynamic cellular behaviors across developmental and disease-related processes. Quantitative benchmarking supports the ability of the framework to recover simulated spatial cell-type abundance patterns, while the inferred redistribution and cell-state dynamics should be interpreted as descriptive computational measures rather than direct evidence of physical cell migration or lineage-validated state transitions.

Keywords Spatial transcriptomics · Single-cell RNA sequencing · Spatiotemporal cell-type redistribution · Inferred cell-state dynamics · Deconvolution · Time-series analysis

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Introduction

Cell migration is a critical and tightly regulated biological process that facilitates tissue rearrangement in response to various stimuli [1]. As a model of collective behavior in biological science, it refers to the coordinated and directional movement of individual cells or groups of cells, driven by molecular signals and changes in the tissue microenvironment [2, 3]. At the cellular level, migration is generally classified into two major subtypes: single-cell migration, often studied in vitro, and collective cell migration, which occurs in vivo [1, 4]. These phenomena have broad biological relevance, including tissue regeneration, wound healing, and cancer progression, where cells spread within tissues or different parts of the body [1, 5]. Because of its beneficial and detrimental roles, controlling cell migration in both time and space is crucial for maintaining tissue homeostasis. Migration also plays a fundamental role in processes such as cell turnover, in which dead cells are removed and replaced by the migration of newly formed cells from underlying tissue layers [6, 7]. Although physical cell migration is one important mechanism underlying tissue remodeling, spatiotemporal changes in tissue organization may also arise from cell proliferation, cell death, differentiation, tissue remodeling, and other biological processes.

In recent years, personalized medicine has become increasingly important in disease treatment, particularly for cancers. Studying spatiotemporal cell-type redistribution and its underlying molecular programs provides a valuable approach for investigating cancer progression, identifying therapeutic targets, and improving precision medicine strategies [8]. Cellular behaviors are influenced by distinct transcriptional and signaling states within heterogeneous cell populations, and given the critical role of cellular heterogeneity in disease progression, particularly in cancer, single-cell technologies have become essential for characterizing cellular states, regulatory programs, and tumor microenvironment interactions [9–11]. Characterizing spatiotemporal cell-type redistribution alongside inferred cell-state dynamics at single-cell and spatial resolution may provide complementary insights into tissue remodeling, disease progression, and therapeutic vulnerabilities [12].

Single-cell RNA sequencing (scRNA-seq) enables discrimination between healthy and cancer cells across tumor progression stages, facilitating more precise prognoses, diagnosis, and treatment strategies [9, 13]. However, scRNA-seq lacks spatial information essential for understanding cell behavior within tissues, and tissue dissociation further complicates assay interpretation [10, 14]. Spatial context—including cell location and local signaling—is critical for understanding cell state and function. Thus, integrating scRNA-seq with spatial transcriptomics is essential for reconstructing spatiotemporal cell-type redistribution and cell-state dynamics. This approach enables characterization of transcriptional patterns, tissue neighborhoods, and local disease-associated features [10]. Spatial transcriptomics addresses these limitations by preserving spatial information through gene expression analysis within intact tissues [10, 14].

We first review related works and provide a detailed description of the proposed framework. We further assess the spatial abundance component through simulation-based benchmarking against alternative spatial deconvolution and spatial mapping approaches, using known simulated cell-type compositions as ground truth (see Additional file 3, Supplementary Notes S2 and S3). We then evaluate the framework using two biologically distinct datasets: human glioblastoma organoids (pathological) and embryonic chicken heart (physiological). Although these case studies serve as representative applications, the framework is generalizable to other tissues, developmental systems, and disease models, particularly additional cancer types. Finally, we summarize the main findings, discuss current limitations, and offer future directions for advancing computational analysis of spatiotemporal cell-type redistribution and inferred cell-state dynamics.

Related works

Characterizing spatiotemporal cell-type redistribution within the tumor microenvironment using high-resolution techniques such as single-cell RNA-seq and spatial transcriptomics remains a major challenge, as current computational approaches remain limited in reconstructing temporal spatial reorganization from these data [11, 15, 16]. In recent years, efforts have been aimed to model cell behavior using imaging and transcriptomic techniques [10]. Here, we review some of these methods.

Chento Wen et al. developed a multiple-step deep learning-based method named 3DeeCellTracker for tracking cells in 3D time lapse images by integrating multiple techniques. Their methods tracked 90–100% of the cells in worm's brain, in a zebrafish heart, and in a cultured tumor. After preprocessing and segmenting 3D + T images into discrete cell regions, the first volume of 3D images manually was corrected, then by applying FFN (feedforward network) + PR-GLS (Point Registration by preserving Global and Local Structures) algorithm [17] in the subsequent volumes, the position of cells was tracked and updated based on spatial patterns of cells preserved between prior and current images in the final, though it does not account for molecular signatures or transcriptome-based cell state transitions [18]. Hideya Aragaki et al. introduced a package for cell tracking and analysis called Link-based Image Matching (LIM) Tracker for time-lapse live imaging over a long period of time consisting three functions, link-type tracking, sequential tracking, and manual tracking, which can be integrated to improve the efficiency of tracking. Moreover, DL recognition function can identify highly accurate processes for a wide range of objects, but its application is constrained to visual modalities without transcriptomic data [19].

Chengxin Li et al. developed an algorithm called Single Cell Track (SC-Track) to raise the accuracy of single cell lineage tracking in time-lapse microscopy images. Traditional CNNs (Convolutional Neural Networks) are prone to produce noisy cell segmentations and classifications. To address this challenge, a hierarchical probabilistic cache-cascade model is employed that leverages biological insights into cell division and movement dynamics [20]. This approach enables SC-Track to outperform existing cell trackers, ensuring robust performance under varying imaging conditions. Additionally, SC-Track includes a function to improve the accuracy of cell classifications in multiclass segmentation time-series. These features make SC-Track a reliable tool when dealing with noisy data from CNN predictions, yet it lacks integration with transcriptomic profiles [20]. Wei et al. proposed CellTrek for mapping single cells to spatial coordinates by integrating scRNA-seq and spatial transcriptomics data. CellTrek enables direct cell-level localization through co-embedding, random forest modeling, and mutual nearest neighbor linking. It also includes the SColoc module for spatial colocalization and SCoexp for detecting spatial co-expression modules. CellTrek has demonstrated superior performance in reconstructing cellular topologies and resolving cell states in various tissues, including the mouse brain, kidney, and DCIS tumors. However, CellTrek primarily addresses spatial localization and mapping of individual cells rather than reconstruction of temporal changes in inferred cell-type abundance across independently sampled spatial time points [11]. Similarly, reference-based deconvolution methods such as CARD [21], RCTD [22], SpatialDecon [23], and SPOTlight [24] estimate cell-type composition or abundance within spatial locations, providing an important basis for downstream spatial analyses. These methods differ in their modeling assumptions, spatial resolution, and ability to recover cell-level spatial information. In our framework, CARD is used as the primary deconvolution method to generate spatial cell-type abundance maps, which are subsequently integrated with CARD_SCMapping for cell-level spatial assignment [21].

Sequence-based approaches provide high-dimensional molecular information, whereas imaging-based approaches provide direct spatial and temporal information at the cellular level. Their respective strengths therefore depend on the biological question, imaging resolution, molecular depth, and experimental design [25–29]. By integrating high-dimensional single-cell RNA-seq with spatial transcriptomics, our framework quantitatively characterizes spatiotemporal cell-type redistribution and inferred cell-state dynamics through complementary spatial and temporal descriptors. Imaging-based methods provide spatiotemporal information for cell tracking, but with limited molecular resolution, and also cannot present migratory heterogeneity of transcriptome-driven

state transitions. In contrast, sequence-based methods such as CellTrek [11] enable high-dimensional molecular profiling and spatial mapping, but they are static and cannot reconstruct temporal changes in inferred cell-type organization at single-cell resolution. Thus, the current methods either determine where cells are and how they move without showing their molecular activity, or reveal the molecular states cells without modeling their motion over time [10].

To address this methodological gap, we developed a computational framework that integrates single-cell RNA sequencing with spatial transcriptomics data to reconstruct temporal changes in inferred spatial cell-type organization and to characterize inferred cell-state dynamics across consecutive time points. Rather than directly tracking physical cell movement or lineage transitions, the framework summarizes temporal tissue remodeling using descriptive spatial and compositional metrics derived from CARD-based spatial abundance estimates and CARD_SCMapping-based cell assignments. Importantly, the proposed framework is complementary to, rather than a replacement for, experimental cell-tracking approaches or existing spatial mapping methods. To quantitatively assess the upstream spatial abundance component of the framework, we additionally benchmarked CARD against alternative spatial deconvolution and mapping methods using simulated data with known cell-type compositions (see Additional file 3, Supplementary Notes S2 and S3). This analysis provides quantitative assessment of spatial abundance estimation while not constituting direct validation of physical migration or lineage transitions.

Methods

In our study, we developed a computational framework to reconstruct spatiotemporal cell-type redistribution and inferred cell-state dynamics by integrating single-cell RNA-seq (scRNA-seq) and spatial transcriptomics (ST) data using CARD (Conditional AutoRegressive-based Deconvolution) [21]. CARD is a reference-based Bayesian deconvolution framework that estimates the relative abundance of annotated cell types at each spatial location while explicitly modeling spatial correlation among neighboring spots. We chose CARD because it provides an interpretable probabilistic framework that preserves local spatial structure and is suitable for moderately sized datasets [21, 30]. Although methods such as Tangram [31], Cell2location [32], and SpaGCN [33] have demonstrated strong performance in specific spatial mapping or deconvolution applications, their computational requirements and methodological assumptions differ from those of the present study. Specifically, Tangram and Cell2location are designed primarily for probabilistic alignment or spatial mapping of single-cell reference profiles to spatial data, whereas SpaGCN emphasizes graph-based spatial domain characterization; in contrast, CARD explicitly models spatial correlation among neighboring spots while estimating cell-type abundance, which better matches the abundance-based input required by our downstream spatiotemporal analysis. Given the objectives and scale of the present study, CARD provided an appropriate balance between biological interpretability, computational efficiency, and the requirements of the present framework [21, 30]. In the simulation-based benchmark, CARD showed intermediate quantitative performance among the evaluated methods, providing quantitatively supported spatial cell-type abundance estimates while offering the spatially informed deconvolution and single-cell mapping capabilities required by the downstream framework (see Additional file 3, Supplementary Notes S2 and S3). Importantly, CARD was selected as the reference-based deconvolution component of the framework rather than as a method for direct temporal cell tracking or lineage inference. The spatial abundance estimates generated by CARD constitute the primary input for the downstream spatiotemporal redistribution analyses described next.

An overview of the proposed framework is shown in Fig. 1. In the first step, single-cell RNA sequencing data are analyzed to identify and annotate transcriptionally distinct cell populations (Fig. 1a). Second, spatial transcriptomics data are quality controlled, normalized, and analyzed to characterize tissue architecture (Fig. 1b). Third, both data modalities are integrated through CARD [21] to reconstruct spatial cell-type abundance maps and infer single-cell spatial assignments through CARD_SCMapping (Fig. 1c). Finally, these reconstructed spatial representations are used to quantify spatiotemporal cell-type redistribution, using abundance-weighted

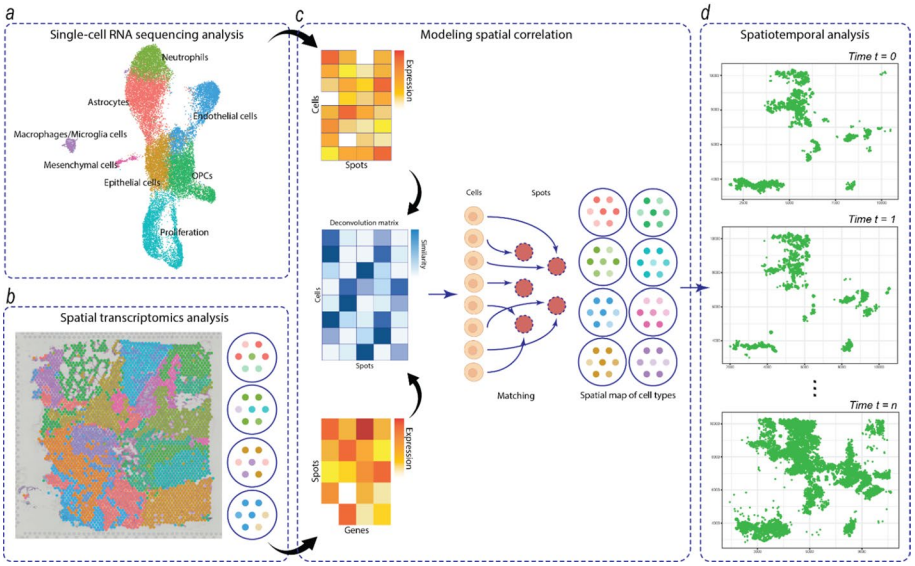


Fig. 1 Main steps of the developed method. **a** Single-cell RNA sequencing analysis. **b** Spatial transcriptomics analysis. **c** Modelling spatial correlation. **d** Spatiotemporal analysis

centroid-based spatial descriptors, and inferred cell-sate dynamics using temporal changes in dominant cell-type assignments across consecutive time points (Fig. 1d). The following sections provide a detailed description of each component of the framework.

Single-cell RNA sequencing analysis

Single-cell RNA sequencing analysis provides high-resolution transcriptional information that enable characterization of cellular heterogeneity and molecular cell type identities [13]. To characterize the cellular composition underlying the spatial analyses, we performed scRNA-seq analysis of human glioblastoma organoids (GBOs) (GSE141946) (see Additional file 1) and embryonic chicken heart tissue (GSE149457) (see Additional file 2). Additional files 1 and 2 summarize the number of biological samples, spatial spots, CARD_SCMapping-inferred cells, the mean numbers of detected genes per cell for each sample and time point, and cell-type-specific abundance estimates.

We used Seurat v5.5.0 [34] to analyze scRNA-seq of glioblastoma tumors samples and corresponding GBOs. Cells with > 400 and < 5000 unique genes were retained, and cells with > 25% mitochondrial gene expression were filtered out. Additionally, genes associated with mitochondrial function, ribosomal proteins, hemoglobin, and MALAT1 were removed from the feature matrix prior to downstream analysis. We merged one primary tumor sample with its two organoid-derived samples at different time points for each patient and then applied standard preprocessing workflows [9, 35, 36], comprising quality control (QC), doublet prediction using scDblFinder [37], normalization, cell-cycle scoring and regression, scaling, and dimensionality reduction using Uniform Manifold Approximation and Projection (UMAP) based on selected principal components (PCs). Cell-cycle scores were computed for each cell using canonical S- and G2M-phase gene signatures, and the difference between these scores (CC.Difference) was regressed out during scaling, together with mitochondrial, ribosomal, and hemoglobin gene percentages and the number of detected genes to reduce potential technical and cell-cycle-related confounding. Cells identified as doublets were excluded prior to downstream analysis; doublet proportions varied across patients (e.g., 6800 cells were identified as doublets for patient UP-8167). Variations between experimental runs including differences in sample preparation, sequencing batches, or library construction, can generate batch effects [35]. Correcting these batch effects is an important consideration when integrating data from multiple experiments. In this study, batch correction was performed using canonical correlation analysis (CCA) implemented in Seurat. CCA aligns cells within a shared correlation space and was used to facilitate integration across samples while retaining biologically relevant transcriptional structure. Finally, clustering was performed using

the Louvain graph-based clustering algorithm implemented in Seurat, and distinct transcriptional clusters were identified for each sample (Fig. 2). Principal-component selection was based on an elbow-plot assessment using two predefined criteria: the first PC at which cumulative explained variance exceeded 90% while the individual contribution was below 5%, and the last PC for which the decrease in variance contribution between consecutive PCs exceeded 0.1%. The minimum of these two values was selected as the number of PCs used for downstream analysis. Clustering resolution was evaluated at 0.5, 0.8, and 1.0, and a resolution of 0.5 was selected based on the biological interpretability and stability of the resulting clusters. These evaluations were used to select analysis parameters within the present datasets; they should not be interpreted as a comprehensive parameter-sensitivity analysis of the entire framework. Differential gene expression was performed by using the FindAllMarkers function in Seurat with a log fold-change threshold of 0.25 and a minimum percent expressed threshold of 0.1, corresponding to 10% of cells expressing the gene. Markers were further filtered to retain genes with an average log fold-change > 0.5 and an adjusted p-value < 0.05. The top 30 marker genes per cluster, ranked by adjusted p-value, were used to assist cell-type annotation. Cell-type identities were assigned based on canonical marker-gene expression, using the Human Protein Atlas [38] as a reference for general cell-type markers together with previously published single-cell transcriptomic datasets from glioblastoma studies [36]. For example, in the UP-8167 parental tumor and corresponding GBOs, we identified eight cell types, including oligodendrocyte precursor cell (OPC)-like, astrocyte-like, proliferation-like, neutrophils, endothelial cells, epithelial cells, macrophages/microglia cells, and mesenchymal cells (Fig. 2a). The same preprocessing and integration workflow was applied to the embryonic chicken heart single-cell transcriptomic dataset with modifications to differential expression analysis as described below. Differential gene expression analysis for the embryonic chicken heart was performed using the FindAllMarkers function with a log-fold change threshold of 0.5. The analysis identified twelve distinct cell types including macrophages, cardiomyocytes, TMSB4X-high cells, immature myocardial cells, vascular endothelial cells, epi-mesenchymal cells, erythrocytes, mural cells, endocardial cells, fibroblasts, valve cells, and epi-epithelial cells (Fig. 2b).

Spatial transcriptomics analysis

The organization of cells in tissue provides essential information about cell states, and the relationships between cells and their environment. Moreover, a cell's location determines the signaling cues it receives, influencing its behavior and function [10]. Spatial transcriptomics has been extensively applied to investigate the tumor microenvironment in cancers, providing critical insights into cellular organization and tissue architecture [13]. In sequencing-based spatially resolved transcriptomics (SRT), gene expression at each spatially defined spot often originates from a heterogeneous population of cells, which may not necessarily belong to the same cell type [39].

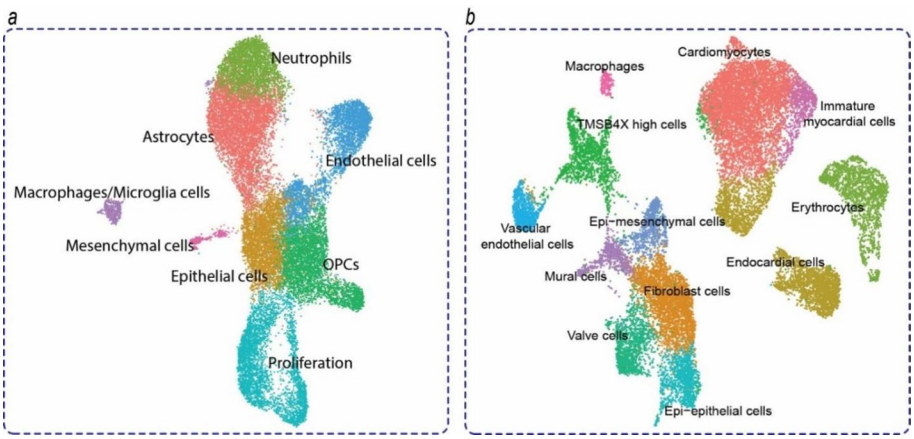


Fig. 2 UMAP visualization of single-cell RNA-seq data. **a** UP-8167 parental tumors and corresponding GBOs. **b** Embryonic chicken heart development sample

Here, we used Seurat, one of the widely utilized computational tools for spatial transcriptomics data analysis [40, 41]. Spatial transcriptomics analysis involves mapping gene expression data to spatial coordinates within intact tissue sections, thereby facilitating the exploration of tissue organization, spatial transcriptional patterns [41]. Our spatial transcriptomics workflow using Seurat can be broadly divided into two primary phases: preprocessing and downstream analysis. The preprocessing phase consist of filtering and preparing high-quality spatial data for subsequent analysis. This phase comprised quality control (QC) and normalization: QC involved removing low-quality spots or features, while the normalization adjusted for differences in sequencing depth and gene expression distributions [9, 10]. In the downstream analysis phase, principal component analysis (PCA) was used for dimensionality reduction, followed by Louvain graph-based clustering implemented in Seurat to identify transcriptionally distinct spatial regions or spot populations. The resulting spatial clusters were visualized using UMAP based on the selected PCs (Fig. 3a) [14]. These analyses were used to characterize the spatial transcriptomic structure of the tissue and to provide the input for subsequent CARD-based deconvolution [21]. Cell-type-specific spatial abundance was not assigned directly from Seurat clustering; instead, cell-type composition at each spatial location was inferred in the subsequent CARD deconvolution step using the scRNA-seq reference. The same spatial transcriptomics preprocessing and analysis workflow was applied to the embryonic chicken heart developmental dataset (Fig. 3b).

Modeling spatial correlation

While scRNA-seq provides gene expression profiles at single-cell resolution, it loses spatial organization due to tissue dissociation [39]. Sequencing-based spatially resolved transcriptomics (SRT) retains spatial information, capturing gene expression from heterogeneous cell populations. By integrating scRNA-seq with spatial transcriptomics, the spatial abundance of diverse cell types can be inferred, enabling the study of tissue architecture and spatially regulated processes such as spatiotemporal cell-type redistribution [10, 14, 41]. This step facilitates spatial characterization of cell-type abundance and provides the basis for subsequent temporal analyses of tissue organization.

To spatially resolve cell populations, we integrated the time-series scRNA-seq data of the first step (i.e., single-cell RNA sequencing analysis) with the spatial transcriptomics data of the second step (i.e., spatial transcriptomics analysis) using the CARD reference-based deconvolution method [21, 30]. CARD uses scRNA-seq data as a reference to estimate the relative abundance of annotated cell types at each spatial spot while incorporating spatial correlation in cell-type composition across tissue locations. We used default CARD parameters for deconvolution and refined spatial mapping with 4000 interpolation grids and 10 spatial neighbors, a minimum gene count of 100 per gene in the reference, and a minimum UMI count of 5 per spatial spot. These settings were used

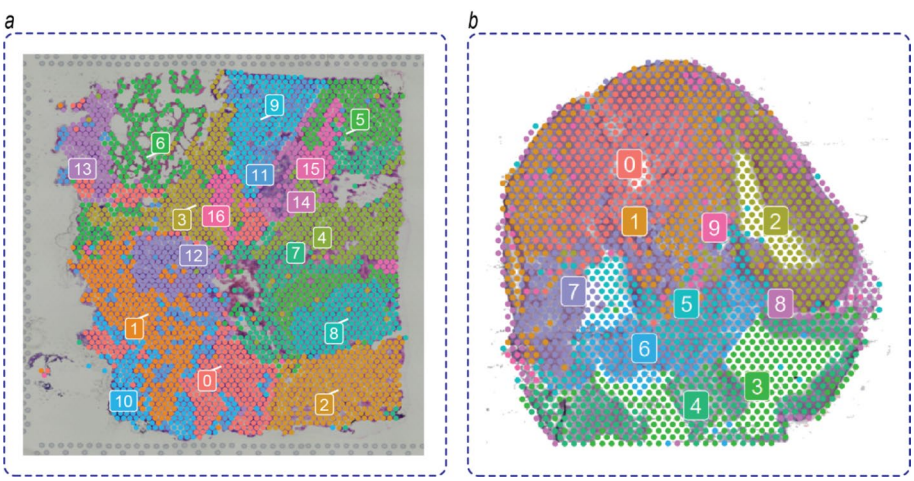


Fig. 3 Spatial transcriptomics analysis. **a** Spatially resolved cell-type localization in human glioblastoma organoids (GBOs). **b** Spatial organization of developmental cell populations in embryonic chicken heart

consistently across the analyzed datasets to provide a common basis for downstream spatial characterization. Ideally, a scRNA-seq reference from the exact same tissue, containing all expected cell types, should be used. Since such a reference was not available in our dataset, we considered each spot in the spatial data as a weighted mixture of the cell types identified by scRNA-seq and applied CARD deconvolution to estimate the cell-type composition at each spot [21]. The resulting spot-by-cell-type abundance matrix represents the primary spatial output of the deconvolution step and was used for subsequent analyses. Cell-type prediction scores for spatial transcriptomes were used to estimate the abundance of each specific cell types at each time point. To identify the spatial distribution of inferred cell-type abundance, spots were labeled according to the cell type with a maximum prediction score and visualized on H&E-stained images (Fig. 4). As shown in Fig. 4, each row represents a distinct time point and shows the inferred spatial abundance distributions of major cell types. These distributions vary across time points, reflecting temporal changes in inferred tissue composition and spatial organization.

These differences may reflect biological processes such as changes in cell abundance, tissue remodeling, or changes in inferred cell-state composition, but may also be influenced by technical and sampling-related factors. We interpret these spatial differences as inferred changes in cell-type organization rather than direct evidence of physical cell migration or experimentally validated cell-state transitions. All procedures outlined in this subsection were uniformly applied to the 14-day time point spatial data from the embryonic chicken heart tissue, as this time point contains both scRNA-seq and spatial transcriptomics data for integration. The resulting spatial abundance maps were subsequently used as the reference spatial representation for the downstream temporal analyses. These refined spatial maps provide a spatially resolved representation of inferred cell-type composition and serve as the primary input for modeling of spatiotemporal cell-type redistribution [42]. Importantly, the downstream framework does not directly track individual cells or estimate physical migration rates. Instead, temporal changes in inferred cell-type abundance distribution are summarized using abundance-weighted spatial descriptors, including centroid shift, abundance-weighted centroid distance (AWCD), and dispersion change. In parallel, inferred cell-state dynamics are characterized by comparing dominant cell-type assignments at spatially corresponding locations across consecutive time points.

Spatiotemporal analysis

As described above, CARD was used to estimate the spatial abundance of cell types across tissue sections sections by integrating single-cell RNA sequencing and spatial transcriptomic data. Building upon deconvolution results, obtained from the third step of the proposed framework (i.e., modeling spatial correlation), we characterized spatiotemporal cell-type redistribution based on changes in the inferred spatial abundance distributions across consecutive time points.

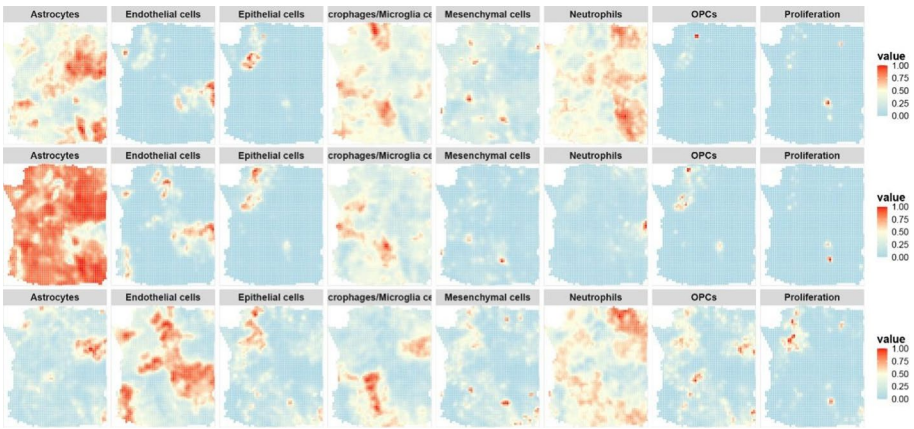


Fig. 4 Visualization of cell-type deconvolution at an enhanced resolution across three time points in UP-8167-GBOs

Spatiotemporal cell-type redistribution was quantified at the cell-type level, rather than at the individual-cell level, using three complementary descriptors that summarize temporal changes in the inferred spatial abundance distribution of each cell type. For each cell type and each time point, an abundance-weighted spatial centroid was computed from the inferred spatial abundance distribution. Centroid shift quantified the Euclidean displacement of the abundance-weighted centroid, thereby capturing the overall change in the spatial location of the inferred cell-type abundance distribution between consecutive time points. Abundance-Weighted Centroid Distance (AWCD) quantified the abundance-weighted average distance of spatial locations from the abundance-weighted centroid, providing a measure of the spatial extent of the inferred cell-type distribution. Dispersion Change quantified the temporal difference in the spatial spread of the inferred abundance distribution around its centroid, indicating whether the inferred distribution became more spatially concentrated or more dispersed over time. Together, these descriptors provide complementary information regarding temporal changes in the location, spatial organization, and dispersion of each inferred cell-type distribution.

To obtain an approximate single-cell spatial representation for the complementary analysis of inferred cell-state dynamics, we applied CARD_SCMapping, an extension of the CARD framework [21], which probabilistically assigns individual cells to spatial coordinates within each spatial spot according to the default mapping procedure implemented in the CARD package [21]. This mapping procedure was performed independently for each temporal sample. For each inferred cell, this procedure yields an estimated spatial coordinate, the coordinate of its corresponding spatial spot, and its assigned cell type. These assignments were subsequently used to characterize temporal changes in inferred dominant cell-type identity at spatially corresponding locations.

Inferred cell-state dynamics were quantified independently of the spatiotemporal cell-type redistribution analysis. At spatially corresponding mapped locations, dominant cell-type assignments were compared across consecutive time points. Locations exhibiting a change in their dominant inferred cell-type identity between consecutive time points were classified as locations with inferred cell-state changes, and the proportion of such locations was quantified for each cell type and temporal interval. These changes in dominant cell-type assignment should not be interpreted as direct evidence that individual cells underwent biological state transitions, differentiation, or lineage conversion. Rather, they represent temporal changes in the inferred dominant cell-type composition at spatially corresponding locations.

Figure 5 provides a conceptual overview of the proposed framework. As illustrated in Fig. 5a, spatiotemporal cell-type redistribution describes temporal changes in the inferred spatial organization of cell-type abundance across tissue sections, as estimated from CARD-derived spatial abundance maps. In contrast, Fig. 5b illustrates inferred cell-state dynamics, representing temporal changes in dominant inferred cell-type identity at spatially corresponding mapped locations based on CARD_SCMapping-derived spatial assignments. Although these two analyses are related, they capture complementary characteristics of tissue organization: redistribution reflects changes in the spatial arrangement of inferred cell-type abundance, whereas inferred cell-state dynamics describe temporal changes in inferred dominant cell-type identity across corresponding spatial locations.

Importantly, neither analysis should be interpreted as direct evidence of physical cell migration, lineage-confirmed differentiation, or experimentally observed cell trajectories. Both analyses represent descriptive computational measures derived from CARD-based spatial abundance inference and CARD_SCMapping-based spatial assignment. Consequently, the observed temporal patterns may reflect the combined influence of multiple biological and technical processes, including physical migration, proliferation, differentiation, cell death, tissue remodeling, sampling variability, differences between independently sampled tissue sections, and uncertainty introduced during spatial deconvolution and cell-type assignment. Accordingly, the framework characterizes temporal reorganization of inferred cell-type abundance and composition rather than directly measuring the biological mechanisms underlying these changes.

To assess the reliability of the spatial cell-type abundance estimates that serve as the upstream input to the spatiotemporal analyses, we additionally performed simulation-based quantitative benchmarking using synthetic multi-timepoint spatial transcriptomic data with known ground-truth cell-type proportions. Multiple spatial deconvolution and single-cell mapping methods were evaluated against the same ground truth using complementary

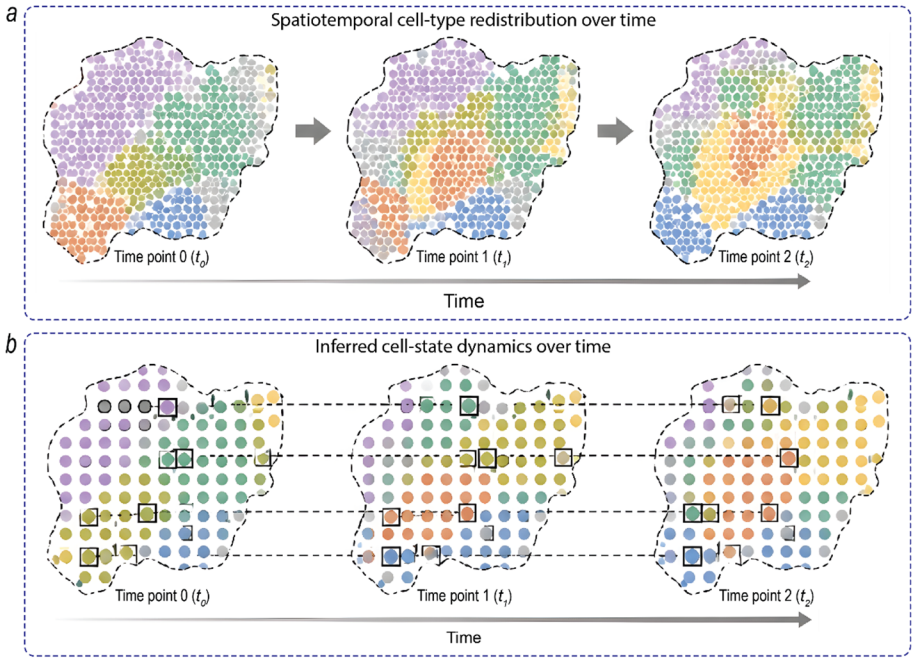


Fig. 5 Conceptual illustration of Spatiotemporal cell-type redistribution and inferred cell-sate dynamics. **a** Temporal changes in the inferred spatial organization and abundance of cell types across successive time points. **b** Changes in dominant inferred cell-type identity at spatially corresponding locations, with dashed lines indicating corresponding locations across time

Table 1 Framework outputs and their interpretation

Framework steps	Results of analysis steps
Single-cell RNA sequencing analysis	Cell-type clusters identified for each sample and time point following QC, doublet removal, batch correction (CCA), dimensionality reduction, and Louvain clustering. Cell identities were assigned using canonical marker-gene expression and visualized by UMAP (Fig. 2)
Spatial transcriptomics analysis	QC-filtered and normalized spatial transcriptomics profiles analyzed using PCA and Louvain community detection to characterize spatial domains and tissue organization, with results visualized by UMAP (Fig. 3)
Modeling spatial correlation	Spot-by-cell-type abundance matrices generated by CARD deconvolution and inferred single-cell spatial coordinates and cell-type assignments obtained using CARD_SCMapping, providing the spatial inputs for the subsequent temporal analyses (Fig. 4).
Spatiotemporal analysis	Two complementary downstream analyses were performed from the CARD and CARD_SCMapping outputs: (i) spatiotemporal cell-type redistribution (Fig. 5a), quantified using abundance-weighted spatial centroids, Centroid Shift, Abundance-Weighted Centroid Distance (AWCD), and Dispersion Change; and (ii) inferred cell-state dynamics (Fig. 5b), quantified from temporal changes in dominant inferred cell-type assignments at spatially corresponding locations.

quantitative metrics (see Additional file 3, Supplementary Notes S2 and S3). Importantly, this analysis evaluates the accuracy of spatial cell-type abundance estimation and does not constitute independent validation of the downstream redistribution or inferred cell-state dynamics.

Framework outputs and interpretative consideration

The primary outputs of the proposed framework are inferred spatial cell-type abundance maps generated at each time point. CARD estimates the relative abundance of each annotated cell type at each spatial location, producing a spot-by-cell-type abundance matrix linked to spatial coordinates. These maps provide the primary input for spatiotemporal cell-type redistribution analysis, whereas CARD_SCMapping provides an approximate single-cell spatial representation used to characterize temporal changes in dominant inferred cell-type identity. Thus, the framework does not directly track individual cells across time but characterizes temporal changes in inferred spatial cell-type organization and dominant inferred cell-type identity using complementary descriptive and statistical measures (Table 1).

Statistical analysis

Statistical analyses were performed independently for each cell type and temporal interval to evaluate temporal changes in inferred spatiotemporal cell-type redistribution and inferred cell-state dynamics. Centroid Shift, Abundance-Weighted Centroid Distance (AWCD), and Dispersion Change were computed for each cell type and consecutive temporal interval and were treated as descriptive measures summarizing distinct aspects of temporal spatial organization. These metrics were not interpreted as direct measurements of physical cell migration.

For inferred cell-state dynamics, spatially corresponding locations were classified according to whether their dominant inferred cell-type assignment remained unchanged or changed between consecutive time points. Fisher's exact test was performed independently for each cell type to determine whether the proportion of locations exhibiting inferred cell-state changes differed across temporal intervals. Because contingency tables occasionally contained small expected counts, Fisher's exact test was selected as the primary inferential procedure. *P* values were adjusted across cell types using the Benjamini–Hochberg false discovery rate (BH-FDR) procedure, and adjusted p-values < 0.05 were considered statistically significant.

Results and discussion

Time-series single-cell transcriptomic data provide a powerful framework for investigating dynamic biological processes by capturing temporal changes in cellular composition and transcriptional programs. However, because spatial transcriptomic measurements are typically obtained from independent tissue sections rather than continuous observations of the same tissue, reconstructing biologically meaningful spatiotemporal patterns requires integrating single-cell transcriptomic information with spatial transcriptomics. To address this challenge, we developed a computational framework that integrates these complementary data modalities to characterize two distinct but biologically related aspects of tissue dynamics: *spatiotemporal cell-type redistribution* and *inferred cell-state dynamics*. The framework was applied to two biologically distinct datasets, including human glioblastoma organoids and embryonic chicken heart development, demonstrating its applicability and generalizability across both disease progression and normal developmental processes.

Figure 6 demonstrates reconstruction of spatiotemporal cell-type redistribution and inferred cell-state dynamics in two biologically distinct systems: human glioblastoma organoids (UP-8167/GBOs) and the developing embryonic chicken heart. In the glioblastoma organoid dataset (Fig. 6a), endothelial cells exhibit progressive temporal redistribution, with the inferred spatial abundance distribution becoming increasingly dispersed over time and extending toward previously unoccupied tissue regions. Inferred cell-state dynamics further reveal temporal changes in dominant inferred cell-type assignments at spatially corresponding locations for neutrophils, macrophages/microglia cells, and astrocytes. These patterns represent temporal changes in inferred cell-type composition rather than direct evidence of lineage transitions, and are consistent with dynamic remodeling of the tumor microenvironment. Similarly, in the embryonic chicken heart (Fig. 6b), valve cells undergo progressive spatial redistribution throughout cardiac development, evolving from a compact spatial distribution into a more elongated tissue organization. Concurrently, inferred cell-state dynamics reveal coordinated temporal changes in dominant inferred cell-type assignments among TMSB4X-high cells, immature myocardial cells, cardiomyocytes, and endocardial cells. These results summarize temporal changes in inferred cell-type composition and are consistent with the progressive spatial organization of major cardiac cell populations during heart development, rather than demonstrating direct lineage relationships. Together, these results demonstrate that temporal tissue remodeling involves both redistribution of inferred cell-type abundance and changes in inferred cell-state composition, highlighting the complementary nature of these two descriptors for characterizing dynamic biological processes. Importantly, these descriptors provide statistical summaries of temporal reorganization inferred from integrated single-cell and spatial transcriptomic data and should not be interpreted as direct evidence of physical cell migration or lineage-validated cell-state transitions. Despite these limitations, integrating single-cell

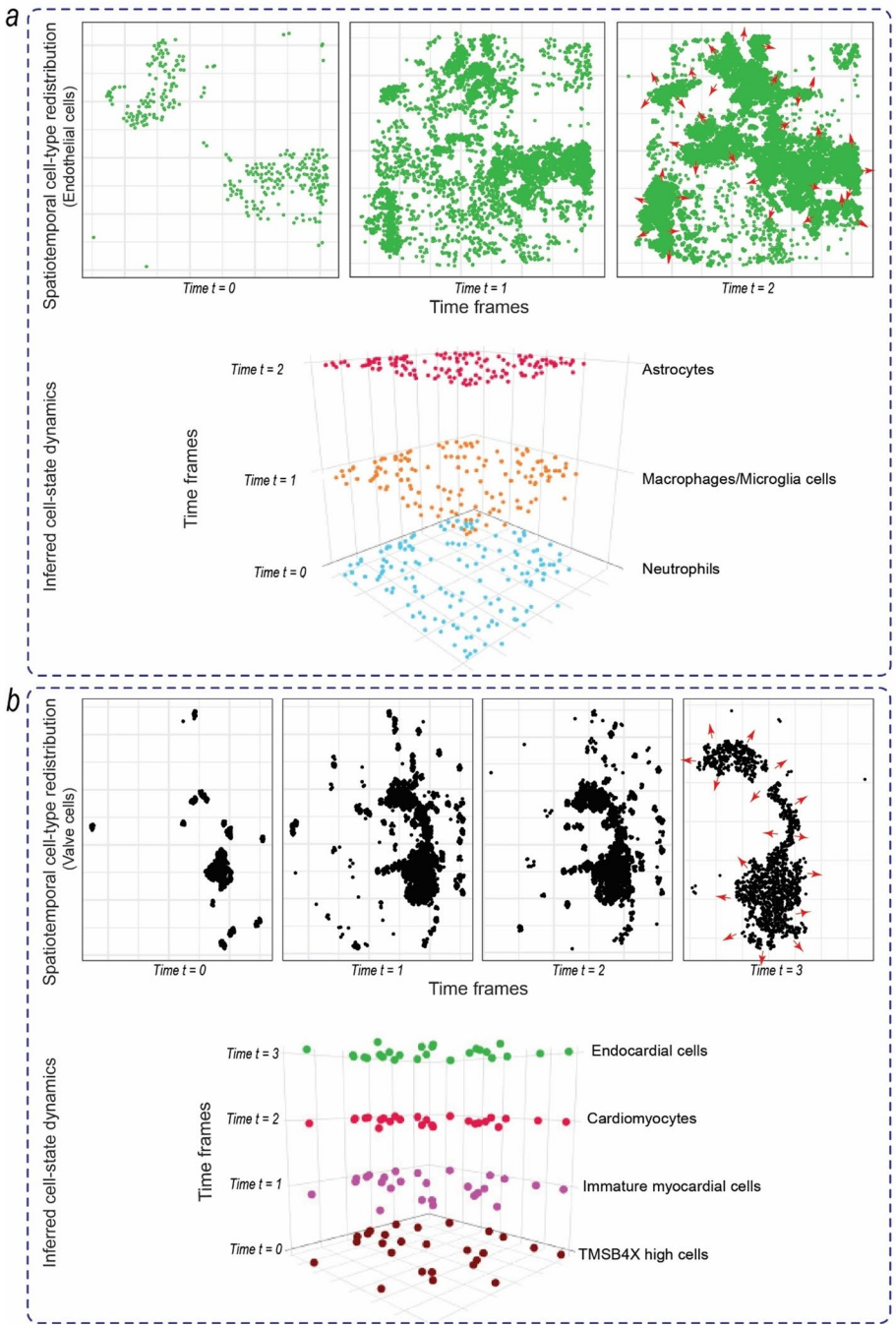


Fig. 6 Spatiotemporal cell-type redistribution and inferred cell-state dynamics in two biological systems. **a** Human glioblastoma organoids (UP-8167/GBOs): endothelial-cell redistribution (upper row) and inferred cell-state dynamics of neutrophils, macrophages/microglia cells, and astrocytes (lower row) across three time points. **b** Embryonic chicken heart: valve-cell redistribution (upper row) and inferred cell-state dynamics of TMSB4X-high cells, immature myocardial cells, cardiomyocytes, and endocardial cells (lower row) across for developmental stages. Red arrows indicate the predominant direction of inferred redistribution at the final time point or developmental stage. Axes represent spatial coordinates, and columns corresponds to individual time points

transcriptomics with spatial transcriptomics substantially extends the biological information obtainable from either modality alone. Rather than relying on a single descriptor, the combination of Centroid Shift, AWCD, Dispersion Change, and inferred cell-state dynamics provides a multi-dimensional characterization of temporal tissue organization. This framework therefore offers a general computational strategy for investigating dynamic spatial biological processes across developmental, physiological, and disease-associated contexts, particularly where longitudinal live-cell measurements are not experimentally feasible.

Dynamic cellular behaviors in glioblastoma organoids (human model)

In the first application of our framework, we analyzed paired single-cell RNA sequencing and spatial transcriptomic data from human glioblastoma tumors and matched glioblastoma organoids (GBOs). Eight major cell populations were consistently identified across the three temporal stages, including neutrophils, macrophages/microglia, OPC-like cells, astrocyte-like cells, proliferation-like cells, endothelial cells, epithelial cells, and mesenchymal cells (Fig. 2a) [36]. The preservation of these major cellular populations across time provided the basis for characterizing temporal changes in both the spatial organization of inferred cell-type abundance and inferred cell-state dynamics. Consistent with the pronounced cellular heterogeneity of glioblastoma, multiple transcriptionally distinct clusters and their corresponding marker genes were recovered, providing the cellular landscape used for subsequent temporal analyses.

Figure 6a demonstrates the reconstructed spatiotemporal cell-type redistribution and inferred cell-state dynamics in glioblastoma organoids. The upper row illustrates the temporal changes in the inferred spatial abundance distribution of endothelial cells across three consecutive time points. The connected centroid positions summarize temporal changes in the abundance-weighted spatial centroid of the inferred endothelial-cell distribution. The progressive displacement of the centroid, together with changes in the spatial extent of the inferred abundance distribution, indicates temporal reorganization of endothelial-cell spatial abundance during organoid progression. The arrows shown at the final time point represent the predominant direction of centroid displacement between consecutive time points. The lower row of Fig. 6a presents inferred cell-state dynamics at spatially corresponding locations across the same temporal series. Representative spatial corresponding locations involving neutrophils, macrophages/microglia cells, and astrocytes illustrate that dominant inferred cellular identities can vary over time while remaining associated with comparable spatial locations. These visualizations should not be interpreted as direct lineage relationships or experimentally validated cell-state transitions. Instead, they represent computationally inferred temporal changes in dominant cell-type assignment following CARD deconvolution and CARD_SCMapping.

Quantitative characterization of spatiotemporal cell-type redistribution is presented in Fig. 7. Three complementary descriptors were evaluated: Centroid Shift quantified the global displacement of abundance-weighted spatial centroids between consecutive time points (Fig. 7a); Abundance-Weighted Centroid Distance (AWCD) summarized the abundance-weighted average distance of inferred spatial abundance from the corresponding centroid (Fig. 7b); and Dispersion Change measured temporal variation in the spatial spread of inferred abundance

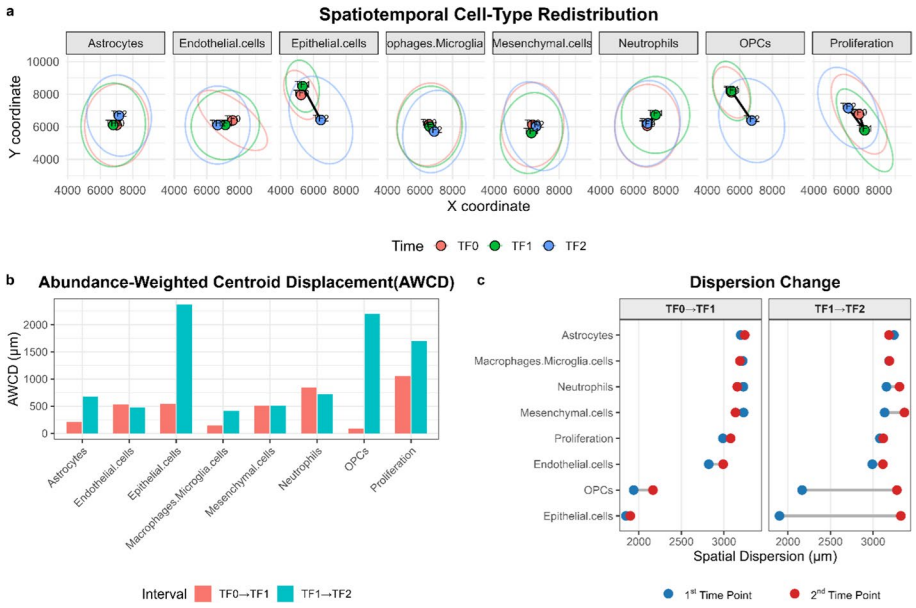


Fig. 7 Quantitative characterization of spatiotemporal cell-type redistribution in human glioblastoma organoids (UP-8167/GBOs). **a** Spatiotemporal redistribution across three consecutive time points ($t = 0 - 2$). **b** Abundance-Weighted Centroid Distance (AWCD) across consecutive temporal intervals. **c** Temporal changes in spatial dispersion of inferred cell-type abundance distributions across consecutive temporal intervals. Blue and red denote the first and second time points, respectively. Each panel shows the dispersion values for a specific time interval

distributions (Fig. 7c). Together, these complementary metrics revealed substantial heterogeneity among glioblastoma cell populations. Endothelial cells exhibited the largest centroid displacement and pronounced changes in spatial organization, whereas macrophages/microglia and OPC-like cells displayed comparatively smaller spatial redistribution. Astrocytes, epithelial cells, and mesenchymal cells showed intermediate redistribution patterns, while neutrophils remained relatively stable across the analyzed temporal intervals. These findings demonstrate that different glioblastoma cell populations exhibit distinct temporal changes in their inferred spatial abundance distributions during organoid development.

Temporal changes in inferred cell-state dynamics were subsequently evaluated using Fisher's exact test with Benjamini–Hochberg false discovery rate correction (Fig. 8). Significant temporal differences in inferred cell-state dynamics were identified for astrocytes, neutrophils, macrophages/microglia, endothelial cells, mesenchymal cells, and OPC-like cells (BH-adjusted $P<0.05$), whereas epithelial cells did not exhibit statistically significant temporal variation (BH-adjusted $P=0.183$). These results indicate that temporal changes in dominant inferred cell-type assignments are cell-type dependent, with some populations exhibiting relatively stable inferred assignments and others showing greater temporal variation across spatially corresponding locations during glioblastoma organoid progression.

Taken together, these analyses demonstrate that glioblastoma organoid development is characterized by coordinated changes in both the spatial organization of inferred cell-type abundance and the composition of dominant inferred cell-type assignments across spatially corresponding locations. The combination of spatiotemporal cell-type redistribution and inferred cell-state dynamics therefore provides complementary perspectives on temporal tissue organization, capturing changes in inferred spatial organization and cell-type composition that cannot be fully described by either analysis alone.

Dynamic cellular behaviors in embryonic chicken heart (animal model)

In the second application of our framework, we analyzed time-series single-cell RNA sequencing and spatial transcriptomic data from the developing embryonic chicken heart. Major cardiac cell populations, including cardiomyocytes, immature myocardial cells, endocardial cells, valve cells, mural cells, vascular endothelial cells, fibroblasts, macrophages, erythrocytes, and epithelial-derived cell populations, were consistently identified throughout cardiac development. Owing to its well-established developmental similarity to the human heart, the

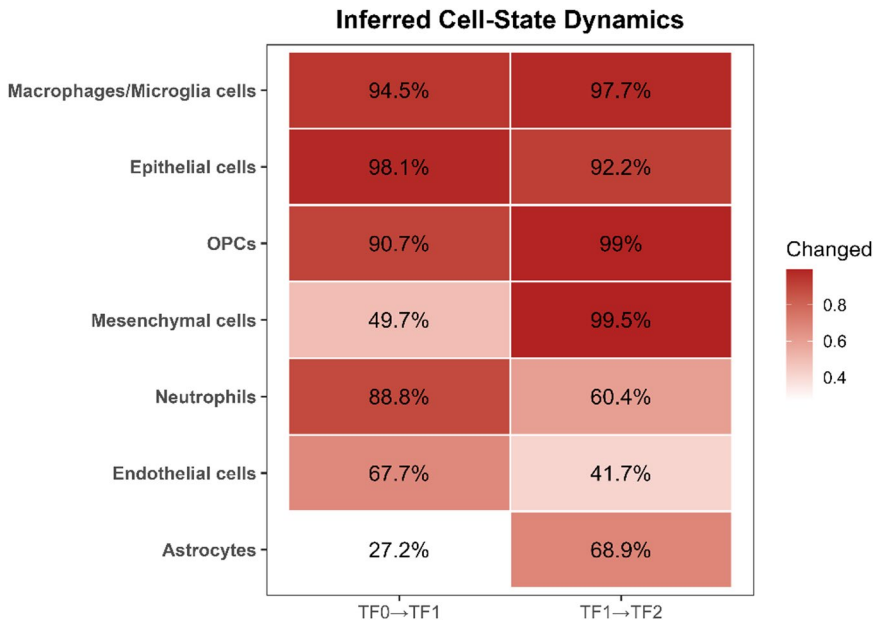


Fig. 8 Inferred cell-state dynamics in human glioblastoma organoids (UP-8167/GBOs). Representative inferred cell-state dynamics across three consecutive time points ($t = 0-2$) based on temporally matched spatial locations. The heatmap shows the statistical significance of temporal differences across cell types using Fisher's exact test with Benjamini–Hochberg false discovery rate correction

embryonic chicken heart provides a suitable model for investigating dynamic changes in tissue organization during cardiogenesis [42].

Figure 6b illustrates the reconstructed spatiotemporal cell-type redistribution and inferred cell-state dynamics throughout four developmental stages. The upper row shows the redistribution of valve-cell abundance across consecutive developmental time points. The temporal changes of abundance-weighted centroid positions summarize changes in the inferred spatial abundance distribution of valve cells. Progressive displacement of abundance-weighted centroids, accompanied by temporal changes in the spatial extent of the inferred abundance distribution, demonstrates temporal reorganization of valve-cell spatial abundance during cardiac development. Red arrows at the final developmental stage indicate the predominant direction of centroid displacement between consecutive developmental stages. The lower row of Fig. 6b presents inferred cell-state dynamics reconstructed from spatially corresponding locations across the same developmental series. Representative cell populations, including TMSB4X-high cells, immature myocardial cells, cardiomyocytes, and endocardial cells, exhibit temporal changes in dominant inferred cell-type assignments while remaining associated with comparable spatial locations. As in the glioblastoma analysis, these results should be interpreted as computationally inferred temporal changes in dominant cell-type assignment following CARD deconvolution and CARD_SCMapping, rather than direct evidence of lineage relationships or experimentally validated cell-state transitions.

Quantitative characterization of spatiotemporal cell-type redistribution is presented in Fig. 9. Three complementary descriptors were evaluated: Centroid Shift quantified the global displacement of abundance-weighted centroids between consecutive developmental stages (Fig. 9a), Abundance-Weighted Centroid Distance (AWCD) summarized the abundance-weighted average distance of inferred spatial abundance from the corresponding centroid (Fig. 9b), and Dispersion Change quantified temporal variation in the spatial spread of inferred abundance distributions (Fig. 9c). These complementary metrics revealed pronounced heterogeneity among cardiac cell populations. Valve cells, TMSB4X-high cells, endocardial cells, and mural cells exhibited comparatively large spatial redistribution during early developmental stages, whereas cardiomyocytes and fibroblasts showed comparatively smaller centroid displacements and more stable spatial organization throughout development.

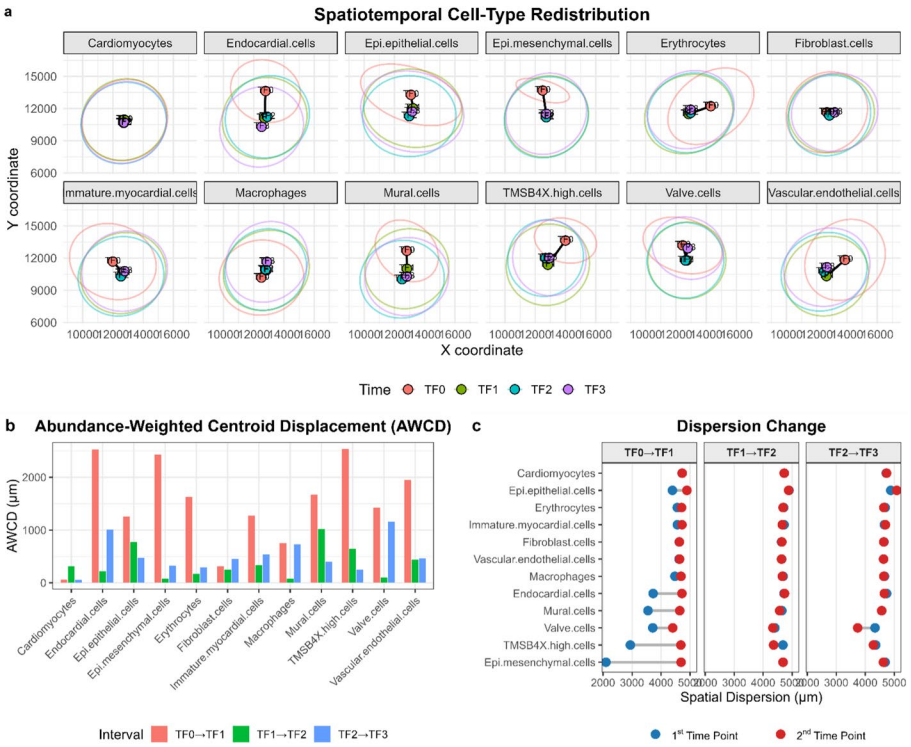


Fig. 9 Quantitative characterization of spatiotemporal cell-type redistribution in embryonic chicken heart tissue. **a** Spatiotemporal redistribution across consecutive developmental stages ($t = 0 - 3$). **b** Abundance-Weighted Centroid Distance (AWCD) across consecutive developmental intervals. **c** Temporal changes in the spatial dispersion of inferred cell-type abundance distributions across consecutive developmental intervals. Blue and red denote the first and second time points, respectively. Each panel shows the dispersion values for a specific time interval

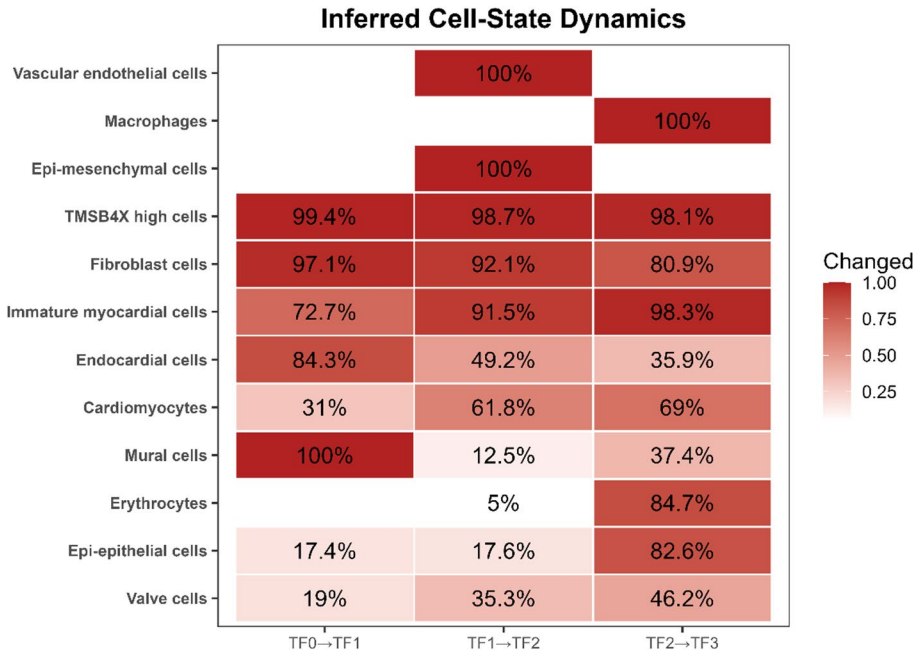


Fig. 10 Inferred cell-state dynamics during embryonic chicken heart development. Representative inferred cell-state dynamics across four developmental stages ($t = 0-3$) based on temporally matched spatial locations. The heatmap summarizes the statistical significance of temporal differences across cell types using Fisher's exact test with Benjamini–Hochberg false discovery rate correction

Collectively, these results indicate that different cardiac cell populations exhibit distinct temporal patterns of inferred spatial reorganization during cardiogenesis.

Temporal differences in inferred cell-state dynamics were subsequently evaluated using Fisher's exact test with Benjamini–Hochberg false discovery rate correction (Fig. 10). Significant temporal variation in inferred cell-state dynamics was detected for cardiomyocytes, TMSB4X-high cells, endocardial cells, valve cells, fibroblast cells, epithelial cells, immature myocardial cells, and erythrocytes (BH-adjusted $P<0.05$). In contrast, mural cells did not exhibit statistically significant temporal variation after multiple-testing correction, whereas macrophages, vascular endothelial cells, and epithelial-mesenchymal cells could not be statistically evaluated because of insufficient observations in one or more temporal intervals. These findings indicate that changes in dominant inferred cell-type assignments vary substantially across cardiac cell populations and occur at different developmental intervals during heart development.

Taken together, the spatiotemporal cell-type redistribution and inferred cell-state dynamics analyses indicate that embryonic cardiac development is characterized by distinct temporal patterns in both the spatial organization of inferred cell-type abundance and the composition of dominant inferred cell-type assignments. The integration of these complementary analyses provides a descriptive characterization of temporal tissue organization, capturing both spatial reorganization of inferred cell-type abundance and temporal changes in inferred cell-type composition throughout cardiogenesis.

Conclusion and future work

This study introduces a framework for characterizing spatiotemporal cell-type redistribution and inferred cell-state dynamics by integrating single-cell RNA sequencing and spatial transcriptomics data across multiple time points. By combining CARD-derived spatial cell-type abundance maps with CARD_SCMapping-based spatial assignments, the framework quantifies temporal changes in inferred spatial organization using complementary descriptors including Centroid Shift, Abundance-Weighted Centroid Distance (AWCD), and Dispersion Change, along with statistical assessment of changes in dominant inferred cell-type assignments.

Application to human glioblastoma organoids and embryonic chicken heart development demonstrated the applicability of the framework across distinct pathological and developmental systems. The framework identified heterogeneous temporal patterns of inferred spatial organization and inferred cell-state composition, illustrating its potential as a general computational strategy for characterizing temporal tissue remodeling from independently sampled spatial transcriptomic datasets.

Simulation-based benchmarking provided quantitative assessment of the spatial cell-type abundance estimates used as input to the downstream analyses. Although CARD did not achieve the highest point-estimation accuracy among the evaluated methods in our benchmark, its spatially informed deconvolution, spatial resolution enhancement, and compatibility with CARD_SCMapping provide a combination of capabilities that is particularly suited to the proposed framework. These characteristics, together with previous studies have reported favorable performance of CARD across several spatial deconvolution benchmarks [21, 30], supported its use as the spatial abundance estimation component of our framework (see Additional file 3, Supplementary Notes S2 and S3). Importantly, neither the benchmark nor the downstream analyses constitute direct validation of physical cell migration, lineage relationships, or experimentally observed cell-state transitions. The inferred patterns should therefore be interpreted as descriptive computational measures of temporal tissue reorganization rather than direct measurements of cellular trajectories or biological state transitions.

Several limitations remain, including dependence on the quality and representativeness of the single-cell reference, spatial sampling and deconvolution uncertainty, and the absence of comprehensive sensitivity analyses and experimental validation of the downstream spatiotemporal metrics. Future studies should evaluate the framework using simulations with explicitly controlled spatial redistribution and cell-state changes, systematic parameter-sensitivity and uncertainty analyses, and experimentally matched longitudinal imaging, lineage-tracing, RNA-velocity, or spatial transcriptomic measurements. Such validation will help establish the robustness and biological interpretability of the proposed framework and extend its application to diverse developmental, disease, and therapeutic contexts.

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Author contributions All authors confirm sole responsibility for the following: study conception and design, analysis and interpretation of results, and manuscript preparation.

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Data availability All data supporting the findings of this study are publicly available. Human glioblastoma tumor and corresponding GBOs data are available through the NCBI Gene Expression Omnibus (GEO) database under accession number GSE141946. The associated spatial transcriptomics data are publicly accessible through the 10x Genomics dataset repository (Human Glioblastoma: Whole Transcriptome Analysis). Embryonic chicken heart spatiotemporal transcriptomic data are available under GEO accession number GSE149457. All data analysis scripts are available on GitHub at https://github.com/pursalimm/tcm.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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Authors and Affiliations

Mahdi Pursalim[1,2] · Kaveh Kavousi[2] · Parisa Shooshtari[3,4,5]

✉ Kaveh Kavousi
kkavousi@ut.ac.ir

✉ Parisa Shooshtari
pshoosh@uwo.ca

1 Department of Bioinformatics, Kish International Campus, University of Tehran, Kish, Iran

2 Laboratory of Complex Biological Systems and Bioinformatics (CBB), Department of Bioinformatics, Institute of Biochemistry and Biophysics (IBB), University of Tehran, Tehran, Iran

3 Department of Pathology and Lab Medicine, Department of Computer Science, Western University, London, ON N6A 5C1, Canada

4 Ontario Institute for Cancer Research, Toronto, ON M5G 0A3, Canada

5 Children's Health Research Institute, London, ON N6C 2V5, Canada